

An Early-Session Sentiment Factor: Timing the Chinese A-Share Market with Intraday Cross-Sectional Breadth

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June 2026

Abstract

I construct and test an **early-session sentiment factor** that measures daily market sentiment from the *cross-sectional breadth* of stock returns in the short window after the open, and uses it to time a long equity book. The economic starting point is a simple, robust fact: information accumulated overnight cannot be traded, and is repriced abruptly on high volume in the first minutes of the session; *breadth*—the fraction of stocks falling—separates a systematic information event from idiosyncratic noise. On point-in-time, survivorship-free A-share data (2015–2025) and under an execution-honest backtest aligned to realistic fill times, early-session breadth predicts the rest-of-day market return (information coefficient $IC = 0.18$, Newey–West $t = 8.1$). Overlaid on a small-cap book, it lifts the full-sample Sharpe ratio from 1.06 to 2.22 and narrows the maximum drawdown from -55% to -32% . A mechanism test shows the factor is not a small-cap-specific alpha but a **broad early-session sentiment beta timer**: it de-risks even large-cap indices, and its payoff scales with the downside risk of the timed asset. I am equally explicit about its boundary: the value is concentrated in bear markets and crashes, it costs return in trending up-markets through opportunity cost (negative in 2025), and it relies on the persistence of early-session weakness into the full day.

Keywords: market timing, market breadth, intraday sentiment, drawdown control, information-compression window

1 Introduction: Motivation and Definition

Chinese small-cap portfolios deliver high long-run returns but carry violent tail risk: the February 2024 micro-cap liquidity cascade drove such books down by more than 50%. If a low-frequency, intraday-executable sentiment proxy can cut exposure when systematic risk arrives, one can shave the tail without giving up most of the upside. This is the objective around which I build the **early-session sentiment factor**.

The economic logic has two layers. First, an **information-compression window**: while the A-share market is closed overnight, global markets, policy, and news keep generating information that cannot be traded; in the first minutes after the open, that information is repriced abruptly and on heavy volume—the most information-dense moment of the day. Second, **breadth, not the index move, is the signal**: the index’s own opening return only reflects large-cap momentum, whereas *how many stocks are falling*—breadth—is what separates a market-wide systematic event from idiosyncratic dispersion. When two-thirds of the market falls together in the opening window, it is a systematic information event, and it tends to persist through the rest of the day via slow information diffusion and liquidity-driven continuation.

Formally, let the early-session return of stock i on day t be $r_{i,t} = P_{i,t}^{09:55}/O_{i,t} - 1$, where $P^{09:55}$ is the 09:55 price and O the day’s open. Define the **down-breadth**

$$D_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}[r_{i,t} < 0], \quad (1)$$

and the timing signal ($1 = \text{hold the long book fully that day}$, $0 = \text{hold cash}$):

$$S_t = \mathbf{1}[D_t \leq \tau], \quad \tau = 0.4. \quad (2)$$

The threshold $\tau = 0.4$ reads: when more than 40% of stocks fall in the opening window, treat the day as systematic risk-off and cut exposure.

2 Data and Backtest Convention

The **universe** spans the Shanghai/Shenzhen Main Board, ChiNext, and STAR Market, excludes the Beijing exchange, and is cleaned point-in-time of ST/*ST names, suspensions, one-way limit locks, and recent IPOs (listed < 250 trading days); **delisted stocks are retained** to remove survivorship bias. The data cover 2015-01 to 2025-12: whole-market opening minute snapshots (09:31/09:45/09:55/10:00) plus daily prices, limit prices, and trading status, about 1.08×10^7 rows. The timed asset is a reproduced equal-weight small-cap book.

The **execution-honest convention** is central. The signal is known only at 09:55, but the full-day return decomposes into a *morning leg* (open to 09:55) and an *intraday leg* (thereafter); empirically about 78% of a cash-day’s fall has already happened by 09:55 and is unavoidable. I therefore fill at **10:00**: on a switching day, only the post-09:55 leg accrues the timing benefit, while the morning-leg loss is borne in full. With pre- and post-switch weights p_{t-1}, p_t and leg returns m_t, r_t ,

$$R_t^{\text{timed}} = p_{t-1} m_t + p_t r_t. \quad (3)$$

Filling at the open (a zero-latency assumption) would overstate the Sharpe by roughly a factor of two. Ability tables below are gross of trading cost; costs are isolated in Section 7.

3 Predictive Power

Motivation. Timing can work only if early breadth genuinely predicts the market return *after* one can trade. I regress the equal-weight market return from 09:55 to close on the continuous predictor $M_t = 1 - D_t$ (up-breadth), with Newey–West (HAC) standard errors.

Evidence. Table 1: IC = 0.18, rank IC 0.20, HAC $t = 8.1$, well beyond the 1% level; the post-signal return on hold days far exceeds that on cash days (+0.36% vs. −0.07% per day).

Conclusion. Early breadth predicts the rest-of-day return with power that survives an auto-correlation correction—the signal carries real information, not fitted noise.

Table 1: Predictive power of early breadth for the post-signal (09:55→close) market return, 2015–2025 ($N = 2674$)

Predictor	IC	Rank IC	t (IID)	t (Newey–West)	Hold vs. cash daily
1–down-breadth	0.176	0.198	9.3	8.1	+0.36% vs. −0.07%

4 Performance and Multi-Window Robustness

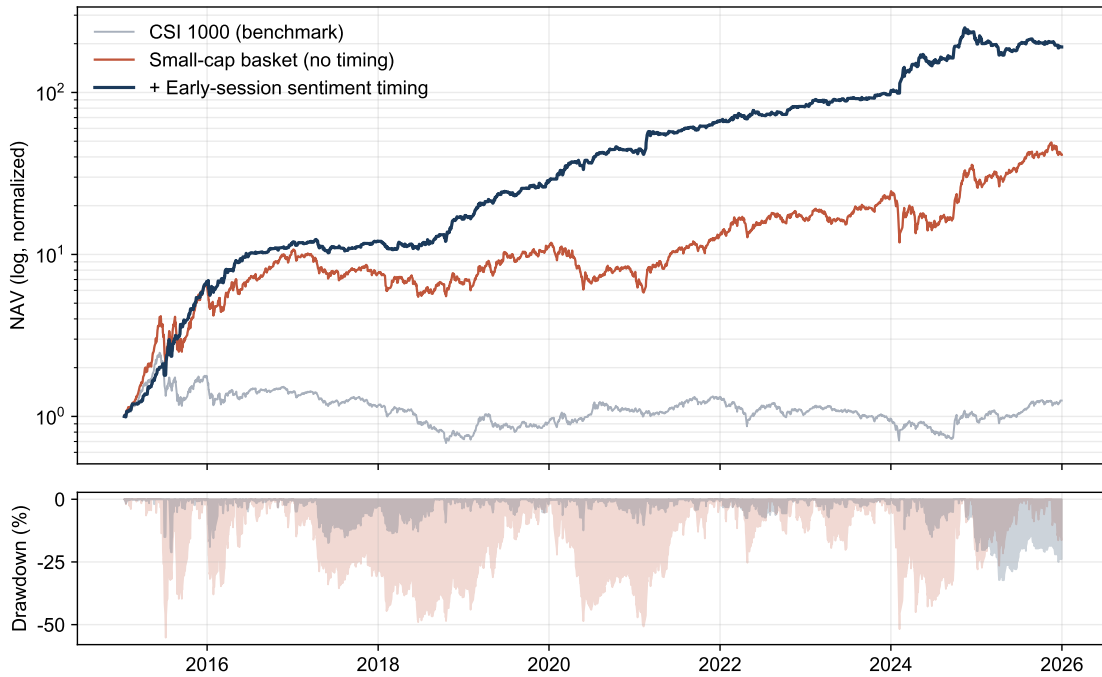
Motivation. Predictive power must translate into a portfolio improvement that does not collapse across market environments. I compare the timed and untimed book over six sample windows, and isolate several stress episodes.

Evidence. Figure 1 plots the small-cap NAV (log) and the underwater drawdown: timing fills in the 2018 bear market and the early-2024 micro-cap cascade. Table 2 and Figure 2 report the full sample, in-sample, out-of-sample, and the trailing three, two, and one years.

Table 2: Sample windows: performance before and after timing (small-cap, execution-honest, gross of cost)

Window	Sharpe		Max drawdown		Annualized	
	Untimed	Timed	Untimed	Timed	Untimed	Timed
Full 2015–2025	1.06	2.22	−55%	−32%	40%	61%
In-sample 2015–2020	1.03	2.79	−55%	−21%	40%	88%
Out-of-sample 2021–2025	1.07	1.48	−52%	−32%	40%	34%
Trailing 3Y 2023–2025	0.94	1.29	−52%	−32%	35%	32%
Trailing 2Y 2024–2025	0.79	1.26	−52%	−32%	30%	36%
Trailing 1Y 2025	1.40	−0.26	−19%	−24%	49%	−7%

Conclusion (including one honest negative result). Timing improves risk-adjusted return and narrows drawdown in the full, in-, and out-of-sample windows. Yet the gain *shrinks over time*, and the **trailing year (2025) is a negative contribution**: small caps trended up that year, and a risk-reducing factor dragged the book from +49% to −7% through opportunity cost. This is not a failure but a direct exposure of the factor’s nature—it is a *risk reducer* that creates value only in systematic downturns, which the next section makes precise.

**Figure 1:** Small-cap NAV (log, top) and underwater drawdown (bottom): timing fills the 2018 and early-2024 tail drawdowns. Benchmark: CSI 1000.

5 Where the Value Comes From: Regimes and Stress Windows

Motivation. If the value concentrates in specific regimes, one must say *where* it is created and *where* it is paid for, rather than hide the structure behind a full-sample average.

Evidence. Classifying bull/sideways/bear by the CSI 2000’s 200-day average and drawdown, Table 3 and Figure 3 show a sharply **asymmetric** payoff: the mean bear-market day turns from −0.057% to +0.155%, while bull and sideways days each give up 0.06–0.17 percentage points to opportunity cost. The three stress windows in Table 4 are starker: through the early-2024 cascade

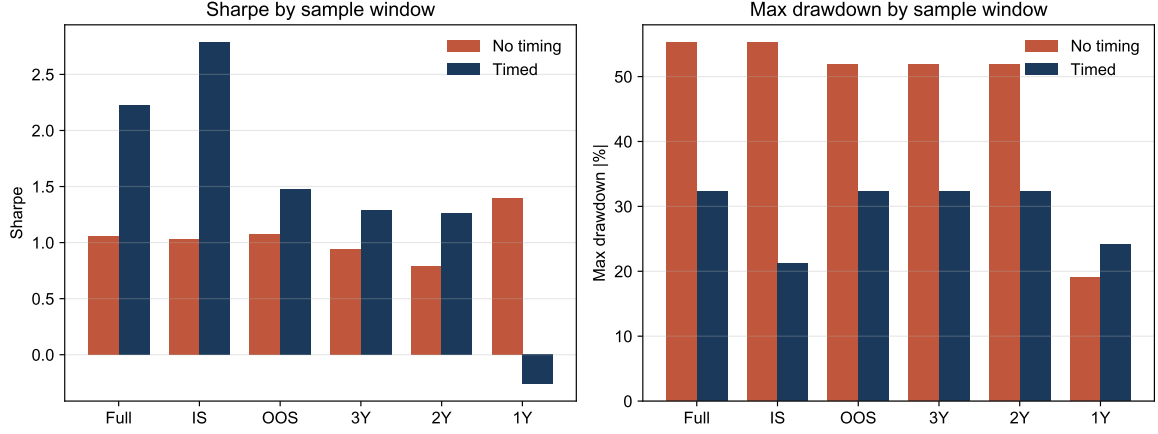


Figure 2: Sharpe (left) and maximum drawdown (right) by sample window, timed vs. untimed. Timing improves both in most windows; the trailing year turns Sharpe negative in a one-sided up-market.

the untimed book returns -45.5% against $+10.7\%$ timed, with drawdown cut from -52% to -5% .

Table 3: Mean daily return by regime

Regime	Days	Untimed	Timed
Bull	1060	+0.211%	+0.148%
Sideways	485	+0.627%	+0.462%
Bear	1124	-0.057%	+0.155%

Table 4: Stress windows: return / max drawdown

Window	Untimed	Timed
2015 crash	+2% / -55%	+190% / -21%
2018 bear	-15% / -30%	+38% / -11%
2024 micro cascade	-46% / -52%	+11% / -5%

Conclusion. The factor is a pure *downside-protection* tool: bear markets and crashes deliver almost all of its value, paid for by a small opportunity cost in bull and sideways markets. This asymmetry is exactly what produced the negative trailing year in Section 4.

6 Economic Mechanism: What It Actually Earns

Motivation. A credible factor needs a falsifiable mechanism, or it cannot be told apart from data mining. I test one hypothesis: is this merely “timing the small-cap size factor”?

Evidence. I apply the *same* early-breadth signal to time four market-cap indices (Table 5, Figure 4). The result is counter-intuitive: even the large-cap CSI 300 benefits (+1.18 Sharpe, 29pp drawdown improvement), so the core is a **broad early-session sentiment beta timer**—it de-risks any long exposure rather than something small-cap-specific. Moreover, controlling for an index’s own opening return, breadth still predicts the rest-of-day return (HAC $t = 3.8 \sim 4.3$), proving it carries cross-sectional information *beyond* large-cap momentum. Drawdown protection strengthens down the cap ladder (CSI 300→1000: $-29 \rightarrow -53$ pp), because small caps fall harder in systematic downturns.

Table 5: The same signal timing indices across cap tiers (execution-honest, gross of cost, 2015–2025)

Index	Untimed Sharpe	Timed Sharpe	Δ Sharpe	Drawdown impr.
CSI 300	0.21	1.40	+1.18	29pp
CSI 500	0.24	1.87	+1.62	46pp
CSI 1000	0.21	2.08	+1.87	53pp
CSI 2000	0.59	2.16	+1.57	24pp

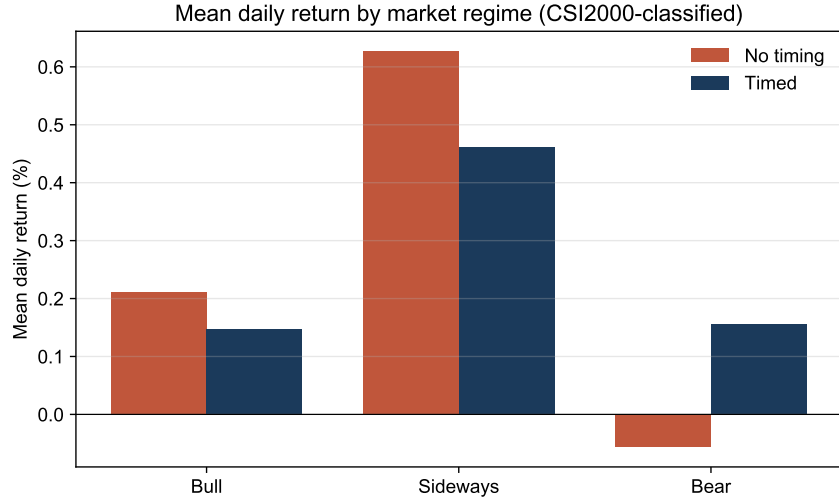


Figure 3: Mean daily return by regime, timed vs. untimed. Only the bear regime is a net positive contribution; bull and sideways carry an opportunity cost.

Conclusion. The factor is a *market-wide* early-session sentiment switch that generalizes across cap tiers, with a payoff that scales with the timed asset’s downside risk. Its failure modes follow from the mechanism: V-shaped intraday reversals (early weakness that does not persist), one-sided bull markets (opportunity cost dominates), and any structural change—more institutional and algorithmic flow, less retail panic—that weakens the persistence of early-session weakness.

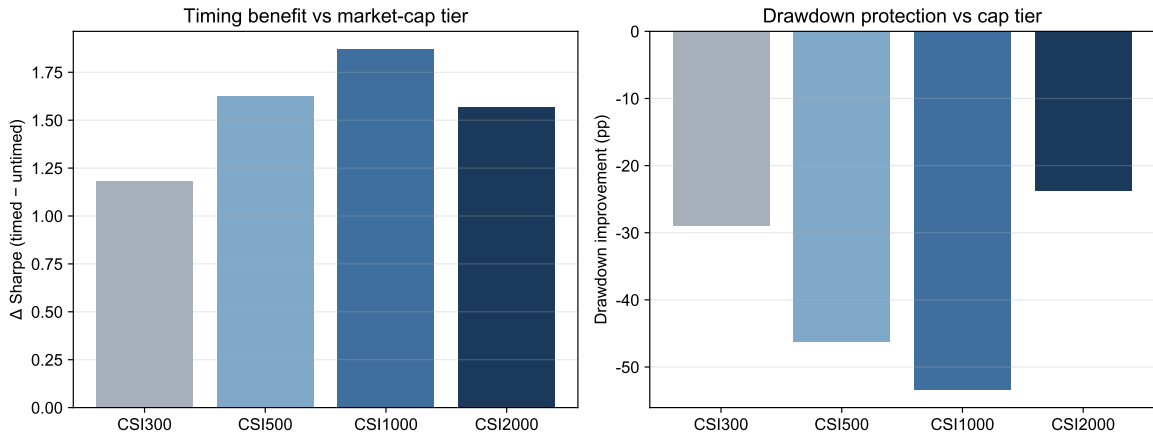


Figure 4: Timing benefit across cap tiers: Δ Sharpe (left) and drawdown improvement (right). Every tier benefits—including the large-cap CSI 300—and protection strengthens down the cap ladder.

7 Execution, Cost, and Robustness

Motivation. The conclusion must withstand four stresses: fill time, transaction cost, observation time, and multiple testing.

Evidence. (i) *Observation time:* at 09:45/09:55/10:00 the timed Sharpe is 1.99/2.22/2.21; 09:45 is hard to compute market-wide in real time, so 09:55 balances information and executability and is the default. (ii) *Cost:* Figure 5 shows the timed Sharpe still reaches 1.74 at a 10bp one-way commission (heavy for micro caps), far above the untimed 1.06. (iii) *Rolling stability:* the rolling one-year Sharpe (Figure 6) is above the untimed book in the large majority of windows. (iv) *Multiple testing:* this line of research ran about 30 signal/parameter variants; against a Bonferroni

threshold (1%, $N \approx 33$) of $|t| > 2.96$, the core factor’s annualized Sharpe corresponds to $t \approx 6.4$, clearing it comfortably.

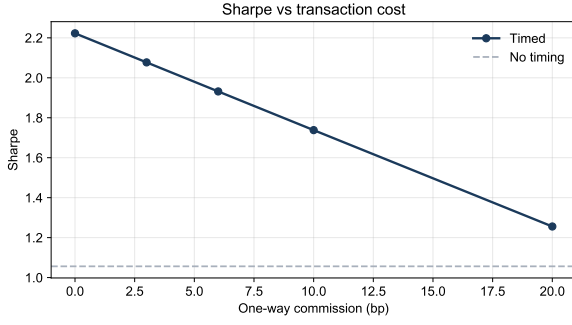


Figure 5: Sharpe decay against one-way cost; dashed line is the untimed benchmark.

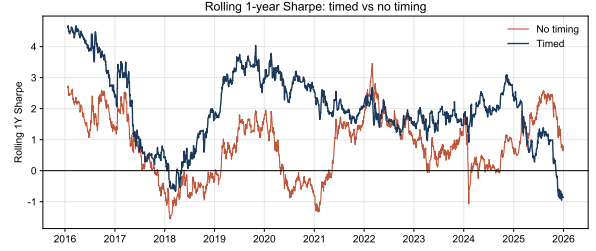


Figure 6: Rolling one-year Sharpe: timed vs. untimed.

Conclusion. The core effect survives cost (to about 10bp), autocorrelation, and a multiple-testing hurdle—a real signal, not an inflated one.

8 Conclusion and Limitations

I have built and tested an **early-session sentiment factor** driven by an information-compression window and centered on cross-sectional breadth. Its value proposition is sharp: *trade roughly two-thirds of the time out of the market for tail protection in systematic downturns*—lifting the small-cap Sharpe from 1.06 to 2.22, cutting the maximum drawdown from -55% to -32% , and flattening the crash troughs in stress windows. The mechanism test shows a market-wide sentiment beta timer that generalizes across cap tiers rather than a small-cap-specific alpha.

Limitations (honest boundary). (i) The value is highly regime-dependent, negative in one-sided up-years (2025), with the out-of-sample gain weaker than in-sample, hinting that efficacy may decay as small-cap tail events become rarer; (ii) results rest on same-convention *relative* differences, and absolute figures are not comparable across accounting conventions; (iii) costs use only a flat-rate stress test, while true micro-cap impact is higher and capacity is not assessed; (iv) a single market and a single asset family—cross-market generalization is untested; (v) real fills are further subject to slippage, for which the 10:00 snapshot is a proxy. Each is left to future work.

A Core Code: Factor and Execution-Honest Backtest

```
# Early down-breadth -> timing signal (observed at 09:55, PIT-cleaned whole market)
df = apply_investable_filter(cache) # main+ChiNext+STAR, ex ST/IPO/limit/halt
df = df[df['open'] > 0]
df['r_m'] = df['price_0955'] / df['open'] - 1 # per-stock early-session return
D = df.groupby('date')['r_m'].apply(lambda x: (x < 0).mean()) # down-breadth D_t
signal = (D <= 0.40).astype(float) # S_t = 1[D_t <= 0.4]

# Execution-honest: signal at 09:55, fill at 10:00; day = morning leg m + intraday leg r
def timed_net(seg, pos, c=0.0, stamp=0.001): # seg: daily m(prev_close->10:00)/r(10:00->close)/d(
    full)
    p = pos.reindex(seg.index).ffill().fillna(1).clip(0, 1)
    pp = p.shift(1).fillna(p.iloc[0])
    gross = pp * seg['m'] + p * seg['r'] # R_timed = p_{t-1}*m + p_t*r
    dp = p - pp
    cost = np.where(dp < 0, -dp * (c + stamp), dp * c) # stamp on sells only
    return (1 + (gross - cost)).cumprod()

# Time-series IC (breadth M=1-D predicting the 09:55->close market return), Newey-West
```

```
import statsmodels.api as sm
res = sm.OLS(fwd.values, sm.add_constant((1 - D).values)).fit()
res = res.get_robustcov_results(cov_type='HAC', maxlags=10) # t_NW = res.tvalues[1]
```

Reproducibility. All factors and backtests are implemented in pure **pandas** with bit-level self-checks; the data are real RiceQuant quotes, cleaned point-in-time and survivorship-free; every figure and number is produced in one pass by a frozen script.